

Business Sales Performance Analytics Report

Introduction

As part of the Future Interns Data Science & Analytics internship, I analyzed a retail sales dataset to understand business performance across products, categories, regions, and customer segments. The objective was to identify key trends, evaluate profitability, and generate actionable insights that could support business decision-making.

The analysis was performed using Python for data cleaning and exploratory analysis, and Power BI for dashboard development and visualization.

Dataset Overview

The dataset contains 9,994 sales records and 21 original features related to orders, customers, products, sales, discounts, and profits.

Key information available in the dataset includes:

- Order details and shipping information
- Product categories and sub-categories
- Customer segments
- Regional and state-level sales data
- Sales, profit, quantity, and discount metrics

After data cleaning, additional features such as Order Year, Order Month, Order Quarter, Shipping Days, and Profit Margin were created to support deeper analysis.

Key Performance Indicators (KPIs)

Metric	Value
Total Sales Revenue	\$2,297,200.86

Total Profit	\$286,397.02
Total Orders	5,009
Profit Margin	12.47%

These KPIs provide a high-level view of overall business performance and served as the foundation for further analysis.

Sales Trend Analysis

Sales performance showed a positive growth trend over the years.

Annual sales revenue:

- 2014: \$484,247
- 2015: \$470,533
- 2016: \$609,206
- 2017: \$733,215

While sales slightly declined in 2015, the business recovered strongly in the following years. The highest revenue was recorded in 2017, indicating consistent growth and increasing customer demand.

Monthly analysis revealed that November and December generated the highest sales, suggesting a strong seasonal effect during the holiday shopping period.

Product Performance Analysis

Among all product categories, Technology generated the highest sales revenue and profit.

Category-wise sales:

- Technology: \$836,154
- Furniture: \$742,000
- Office Supplies: \$719,047

Category-wise profit:

- Technology: \$145,455
- Office Supplies: \$122,491
- Furniture: \$18,451

Technology products not only generated the highest revenue but also contributed significantly to profitability.

The highest revenue-generating product was:

- Canon imageCLASS 2200 Advanced Copier

This product also ranked among the most profitable products in the dataset.

Regional Analysis

Regional analysis highlighted significant differences in performance across locations.

Sales by region:

- West: \$725,458
- East: \$678,781
- Central: \$501,240
- South: \$391,722

Profit by region:

- West: \$108,418
- East: \$91,523
- South: \$46,749
- Central: \$39,706

The West region consistently outperformed other regions in both sales and profit. California emerged as the strongest-performing state and contributed significantly to overall business success.

Customer Analysis

Customer segment analysis revealed that the Consumer segment generated the highest sales volume compared to Corporate and Home Office customers.

Shipping mode analysis showed that Standard Class was the most frequently used shipping option and contributed the highest share of sales.

These observations suggest that the business relies heavily on consumer purchases and standard shipping operations.

Impact of Discounts on Profitability

Discount analysis showed a clear relationship between higher discounts and lower profitability.

Orders with discounts above 40% frequently resulted in negative average profits, while orders with little or no discount generated healthier profit margins.

This indicates that aggressive discounting may increase sales volume but can negatively affect overall profitability if not managed carefully.

Recommendations

Based on the analysis, the following recommendations can help improve business performance:

1. Increase focus on Technology products due to their strong sales and profit contribution.
 2. Expand successful strategies used in the West region to other regions.
 3. Review high-discount sales policies to reduce profit loss.
 4. Prioritize high-performing products in marketing campaigns and inventory planning.
 5. Strengthen customer retention initiatives for the Consumer segment.
 6. Leverage seasonal demand patterns during November and December through targeted promotions.
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Conclusion

This project provided valuable insights into sales performance, profitability, customer behavior, and regional trends. The combination of Python-based analysis and Power BI dashboards helped transform raw sales data into meaningful business insights. The findings can support data-driven decisions related to product strategy, regional expansion, pricing, and customer engagement.